ECON 0100 | Exercise E1 | Costs

Solutions

Answers and work are shown in the red blocks.

Q1 | Economic Sudoku

Use the relationships between cost variables to solve for the missing values in the table.

OUTPUT	AFC	AVC	ATC	MC
1	120	100	220	
				80
2	A =	B =	150	
				90
3	40	90	130	
				130
4	30	C =	D =	
				E =
5	24	108	132	

Fixed cost is the same at every output: $FC = AFC \times Q = 120 \times 1 = 120$ (and 40×3, 30×4, 24×5 all give 120 too).

A = 60. $AFC = FC/Q = 120/2 = 60$.

B = 90. $AVC = ATC - AFC = 150 - 60 = 90$.

C = 100. Total cost at 3 units is $TC = ATC \times Q = 130 \times 3 = 390$. Marginal cost from 3 to 4 units is 130, so $TC = 390 + 130 = 520$ at 4 units. Then $VC = TC - FC = 520 - 120 = 400$ and $AVC = 400/4 = 100$.

D = 130. $ATC = TC/Q = 520/4 = 130$. As a check, $AFC + AVC = 30 + 100 = 130$.

E = 140. Total cost at 5 units is $132 \times 5 = 660$, so $MC = \Delta TC/\Delta Q = (660 - 520)/1 = 140$.

The given marginal costs check out the same way: total cost is 220, 300, 390, 520, 660 at 1 to 5 units, and the steps are 80, 90, 130, 140.

OUTPUT	AFC	AVC	ATC	MC
1	120	100	220	
				80
2	A = 60	B = 90	150	
				90
3	40	90	130	
				130
4	30	C = 100	D = 130	
				E = 140
5	24	108	132	